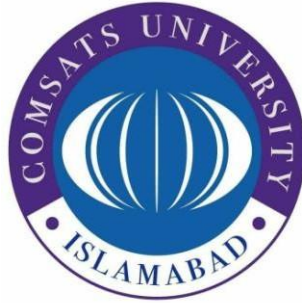


COMSATS University Islamabad

Attock Campus



Department Of Computer Science

<i>Course</i>	<i>INFORMATION SECURITY</i>
<i>Instructor</i>	<i>MISS AMBREEN GUL</i>
<i>Assignment No.</i>	<i>01</i>

Student Details

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CODE:

```
# Function to encrypt the plain text
def caesar_encrypt(text, shift):
    encrypted_text = ""
    for char in text:
        # Check if character is uppercase letter
        if char.isupper():
            # Convert character to ASCII, shift it, wrap around using modulo 26
            encrypted_char = chr((ord(char) - 65 + shift) % 26 + 65)
            encrypted_text += encrypted_char
        # Check if character is lowercase letter
        elif char.islower():
            # Convert character to ASCII, shift it, wrap around using modulo 26
            encrypted_char = chr((ord(char) - 97 + shift) % 26 + 97)
            encrypted_text += encrypted_char
        else:
            # If character is space or special character, keep it unchanged
            encrypted_text += char
    return encrypted_text

# Function to decrypt the cipher text
def caesar_decrypt(ciphertext, shift):
    decrypted_text = ""
    for char in ciphertext:
        # Check if character is uppercase letter
        if char.isupper():
            decrypted_char = chr((ord(char) - 65 - shift) % 26 + 65)
            decrypted_text += decrypted_char
        # Check if character is lowercase letter
        elif char.islower():
            decrypted_char = chr((ord(char) - 97 - shift) % 26 + 97)
            decrypted_text += decrypted_char
        else:
            # Keep spaces and special characters unchanged
            decrypted_text += char
    return decrypted_text

# Taking input from user
message = input("Enter your message: ")
shift_value = int(input("Enter shift value: "))

# Encrypt the message
encrypted_message = caesar_encrypt(message, shift_value)
print("Encrypted Message:", encrypted_message)

# Decrypt the message
decrypted_message = caesar_decrypt(encrypted_message, shift_value)
print("Decrypted Message:", decrypted_message)
```

OUTPUT:

```
C:\Users\DELL\PycharmProjects\informationsecurity\.venv\Scripts\python.exe C:\
Enter your message: HI im haroon khan
Enter shift value: 4
Encrypted Message: LM mq levssr oler
Decrypted Message: HI im haroon khan

Process finished with exit code 0
```

CODE EXPLANATION:

This program implements the **Caesar Cipher**, a basic encryption technique in which each alphabet letter in a message is shifted by a fixed number of positions called the key. The program first defines the function `caesar_encrypt(text, shift)`, which takes a message and a shift value as input and initializes an empty string to store the encrypted result. It then loops through each character of the input text and checks whether the character is uppercase or lowercase using `isupper()` and `islower()`. If the character is uppercase, it converts it to its ASCII value using `ord()`, shifts it forward by the given key, applies modulo 26 to wrap around the alphabet if necessary, converts it back to a character using `chr()`, and appends it to the result string. The same process is applied to lowercase letters using their ASCII base value (97). If the character is a space, number, or special symbol, it is added unchanged to maintain the original format. The `caesar_decrypt(ciphertext, shift)` function works similarly but subtracts the shift value instead of adding it to return the original message. In the main program, the user enters a message and a shift value, the message is encrypted using the encryption function, and then decrypted using the decryption function to demonstrate that the original message can be successfully recovered using the same key.